

Operating System - Class Test MCQs

1. What is an Operating System?

- A) A program that controls hardware only
- B) A program that acts as an intermediary between the user and hardware**
- C) A program that runs only games
- D) A program for web browsing

2. Which of the following is considered the kernel?

- A) The application software
- B) The hardware
- C) The one program running at all times**
- D) The user interface

3. Which type of storage is volatile?

- A) Main memory**
- B) Hard disk
- C) Solid-state disk
- D) Optical disk

4. Which device issues an interrupt?

- A) Mouse
- B) CPU
- C) Device controller**
- D) Keyboard

5. What is a bootstrap program?

- A) A user application
- B) A program that initializes the system**
- C) A gaming software
- D) A program to create backups

6. Which of the following storage is fastest?

- A) Hard disk

- B) Main memory
- ☒ C) Cache memory
- D) DVD

7. Which system uses more than one CPU?

- A) Single processor system
- B) Batch processing system
- ☒ C) Multi processor system
- D) Embedded system

8. Which OS type allows more than one program to run concurrently?

- A) Real-time OS
- B) Batch OS
- ☒ C) Multi-tasking OS
- D) Embedded OS

9. In symmetric multiprocessing, each processor has:

- A) Different OS
- B) No OS
- ☒ C) Same OS
- D) Only main memory

10. What is the function of a device driver?

- A) Runs applications
- B) Manages file system
- C) Manages CPU
- ☒ D) Provides interface between device controller and kernel

11. DMA stands for:

- A) Direct Main Access
- B) Direct Memory Allocation
- C) Data Memory Access
- ☒ D) Direct Memory Access

12. Interrupts can be classified into:

- ☒ A) Two types: software and hardware
- ☐ B) Only hardware
- ☐ C) Only software
- ☐ D) Three types

13. Which is NOT a function of an Operating System?

- ☐ A) Memory management
- ☒ B) Application development
- ☐ C) File management
- ☐ D) Device management

14. Cache memory is located:

- ☐ A) Inside the hard disk
- ☐ B) Near the monitor
- ☒ C) Closer to the CPU
- ☐ D) Outside the motherboard

15. Which OS does NOT require interaction with the user?

- ☐ A) Multi-tasking OS
- ☐ B) Real-time OS
- ☒ C) Batch OS
- ☐ D) Network OS

16. Which system uses jobs grouped into batches?

- ☐ A) Time-sharing OS
- ☐ B) Multi-processing OS
- ☒ C) Batch OS
- ☐ D) Network OS

17. Which of the following is NOT an element of a computer system?

- ☐ A) CPU
- ☐ B) Device controller
- ☐ C) Main memory

☒ D) Mouse pad

18. What is cache coherency?

- A) Ensuring memory is read-only
- ☒ B) Ensuring all CPUs have the most recent value
- C) Ensuring devices don't interrupt each other
- D) Ensuring the OS boots properly

19. Which of these is NOT a type of OS mentioned?

- A) Network OS
- B) Batch OS
- ☒ C) Firmware OS
- D) Real-time OS

20. What is the role of a multiprogramming OS?

- A) Execute only one program
- ☒ B) Execute multiple programs concurrently
- C) Run only background tasks
- D) Install operating systems

21. What is an interrupt?

- A) A user command
- B) A request for printer access
- ☒ C) A signal needing immediate attention
- D) A system crash

22. Who initiates the DMA transfer?

- A) User
- B) Operating system
- ☒ C) CPU
- D) Peripheral

23. What is firmware?

- A) Part of cache

- B) Program stored in RAM
- C) Bootstrap program in ROM/EPROM**
- D) Application software

24. A dual-core design means:

- A) Two separate CPUs
- B) One chip with two cores**
- C) Two keyboards
- D) Two memory modules

25. Time-sharing system is an extension of:

- A) Single tasking OS
- B) Real-time OS
- C) Batch OS
- D) Multiprogramming OS**

26. Main memory is:

- A) Non-volatile
- B) Volatile**
- C) Magnetic
- D) Portable

27. Real-time OS is used in:

- A) Gaming PC
- B) Banking software
- C) Industrial control systems**
- D) Web servers

28. Which of the following interacts directly with the user?

- A) Kernel
- B) Cache
- C) User interface**
- D) ROM

29. Job accounting is used to:

- A) Track passwords
- B) Manage jobs
- ☒ C) Track resource usage
- D) Handle interrupts

30. The function of OS in security is to:

- A) Develop antivirus software
- ☒ B) Protect data from loss and damage
- C) Delete malware
- D) Track network data